THE UNITED REPUBLIC OF TANZANIA

ZAMUPE EXAMINATIONS COMMITTEE

FORM FOUR JOINT MOCK EXAMINATION, AUGUST 2023

032 **CHEMISTRY 1**

(For Both School and Private Candidates)

TIME: 3 Hours **Year: 2023**

Instructions

1. This paper consists of sections A, B and C with a total of **eleven (11)** questions.
2. Answer **all** questions in sections A and B and **two (2)** questions from section C.
3. Section A carries **sixteen (16)** marks, section B **fifty four (54)** marks and section C carries **thirty (30)** marks.
4. Cellular phones and any unauthorised materials are **not** allowed in the examination room.
5. All drawings should be in pencil.
6. Write your **Examination Number** on every page of your answer booklet(s).
7. The following constants may be used Atomic masses;
   H=10=16,Na = 23
   C=12, Cl=35.5, N=14, Zn =65, S=32,
   Ca=40, Fe=56, Cu=64, Al=27
   GMV at S.T.P =22.4dm^3
   Avogadro's number= 6.02x10^{23}
   1 Faraday = 96500 coulombs
   1 Litre = 1dm^3 = 1000 cm^3

wazaelimu.com

SECTION A (16 Marks)
Answer ALL questions in this paper

1. for each of the items (i-x) choose the correct answer from the given alter natives and write it's letter in the answer sheet provided:-

(i) Technicians prefer to use blue flame in welding because.
- A. It is bright and non and non-soot
- B. it is light and no-soot
- C. it is very hot and non-soot
- D. it is not expensive

(ii) One of the isotopes of an element X has atomic number Z and a mass number A. What is the number of Neutrons contained in the nucleus of the element
- A. Z-A
- B. A
- C. A+Z
- D. A-Z

(iii) Substance X liberates chlorine gas from acidified potassium chloride what is the behaviour described by substance X?
- A. Catalysis
- B. Oxidation
- C. reduction
- D. decolourization

(iv) Aluminium does not react with water and does not corrode much in air why?
- A. It below hydrogen in reactivity series
- B. it very stable
- C. it form stable carbonate which prevent reactions
- D. it's covered with protective coating of an oxide

(v) Why pharmacists advices the uses of ant-acids tablets to chew rather than swallowing without chewing?
- A. Chewed tablets easier to swallow
- B. to experience to taste of ant-acids tablets
- C. to slow down the rate of effects of tablets
- D. to increase the surface area to volume ration of the tablets

(vi) When water is added to an acid, the acid becomes
- A. More acidic and it's pH goes down
- B. More acidic and It's pH goes up
- C. less acidic and it's pH goes up
- D. less acidic and it's pH goes down

(vii) Which term describes a rapid chemical reactions that released energy in the form of light and heat
- A. Ignition
- B. combustion
- C. reactivity
- D. heating

(viii) The process of giving away water of crystallization to the atmosphere by a chemical substance is called

A. Deliquescence
B. Efflorescence
C. hygroscope
D. Sublimation

(ix) An electric current was passed through a concentrated solution of hydrochloric acid using carbon electrodes. The substance liberated at anode was

A. oxygen
B. chlorine
C. hydrogen
D. sodium

(x) Butane can occurs in different structures either in a condensed or open structure but remains with four atoms of carbon and ten atoms of hydrogen such occurrence of a molecules is termed as

A. Allotropy
B. Isomerism
C. Isotophy
D. Amphoterisim

2. Match the properties of element in LIST A with the respective element in LIST B by writing the letter of the correct response besides the item number in the answer sheet provided

LIST A	LIST B
(i) A black sold dement which burns with reddish glow giving off colourless gas which slightly acidic	A. Sodium
(ii) Silvery white metal which burns with golden yellow flame giving an oxide which is basic in nature	B. Iron
(iii) A yellow element in colour which bums with blue flame giving colourless gas which is strong acidic in nature	C. calcium
(iv) A silvery white metal which burns with brick red giving off oxide which is white solid	D. lead
(v) A shining white strip metal burns with dazzling white flame giving an oxide which is slightly basic in nature	E. zinc
(vi) It's oxide is brown when hot and yellow when cold	F. carbon
	G. magnesium
	H. sulphur

SECTION B (54 MARKS)

Answer all questions in this section

3. Answer all question in this section

(a) An isotopes of element X has mass number of 31 and an atomic number of 15

(i) Write the name of element x

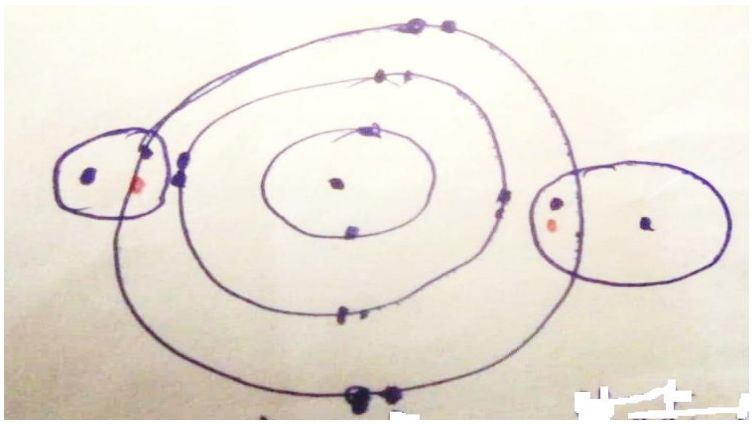
(ii) write its nuclide nutation

(iii) how many neutrons does it have?

(iv) write down the group and period of elements X

(b) The experiment result of J.J. Theomson said that atoms contains charged particle, briefly explain why it's electrically neutral

(c) a molecule of a certain gas can be represent by the following diagram



(i) What type of bonding that holds atoms of the molecule

(ii) name other two compound which the type of bonding

4. (a) Give the name of the metals which fits the following descriptions

(i) It produces cracking sound when burn and it's oxide is brown when hot and yellowish when cold

(ii) it's green coloured carbonate decompose when heated to form a black oxide

(iii) it must be kept under paraffin to protect it from reaction with air

(b) X and Y are two oxide of metals. X is yellow when hot and white when cold, white Y is black when hot and black when cold

(i) Identify oxides x and y

(ii) with the aid of balanced chemical equation, write the observation when compound x reacts with dilute sulphuric acid

(iii) suggest the type of each oxide, x and y

5. Chemistry teacher guided students of form four to study carefully an experiment carried out in order to distinguish various substances formal during an experiment. The procedures were as follows
   (a) Measure about 120cm3 of water and pour it in a beaker
   (b) add a spatula full of common salts and stir to allow salts to dissolve
   (c) continue to add more salts to the solution, stirring until no more salts can dissolve
   (d) place the solution on tripod stand and heat gently as you continue stirring
   (e) stop heating when salts dissolves
   (f) place the beaker in a trough of cold water and allow cooling for 3 to minutes note the results observed
   (i) What type of solution was obtained when a spatularful of table salt were dissolved in 120 cm3 of water at room temperature? Give reasons
   (ii) What type of solution was obtained when no more salts dissolved in cold water at room temperature? Give reason
   (iii) What is the name of the final solution obtained after cooling it through of cold water? Give reason
   (iv) Explain the two (2) application of the concept studied above

6. (a) A person suffering from indigestion produces 1.0 Litre of gastric juices per day which contains about 2.0g of hydrochloric acid. How many antacid tablets each containing 400mmg of sodium bicarbonate (NaHco3) is needed to neutralize all the hydrochloric acid produced in a day?
   (b) Name the suitable warning sign that you would find on the following:-
   (i) A bottle containing concentrated sulphuric in a day?
   (ii) A bag containing rat or insect poison
   (iii) Room door containing x – ray machine
   (iv) Bag containing dynamites

7. (a) Explain how the following factors affect the rate of chemical reaction
   (i) Pressure (ii) Temperature
   (b) Sodium hydroxide was dissolved in water after a while the container. Holding the solution was observed to be hat
   (i) Basing on the heat change, name the reaction that took place in the above process
   (ii) Suggest any two application of the reaction above in our every day
   (iii) Draw energy level diagram for the above process named in b (i)

8. (a) briefly explain by giving three (3) reasons why most of laboratory apparatus are made up made up of glass materials

   (b) Zawadi is a form one student when there was conducting experiment in the laboratory. She fail to handle the beaker that made her injury, you are to provide first Aid to her, give three reasons why you should provide first and to Zawadi

   (c) Name the best method that can be used to separate each of the following substance

   (i) Mixture of petrol and diesel

   (ii) Mixture of paraffin and water

   (iii) Tiny pastilles of iron and wheat

SECTION (30 Marks)

Answer any two question inn this question in this question

9. (a) Analysis of organic compound contains the following pasted 2.84g of carbon combine with 0.47g of hydrogen and 3.7g of oxygen. If half of its vapour density is 15

   (i) work out molecular formula

   (ii) write down its I UPAC name

   (iii) Give the general formula of organic compound in a (i) above

   (b) Write and name all possible Isomer of pentane

10. Agriculture is back home of our economy" but the addition of in organic fertilizer in the farm is not as Important as addition of organic manure. Discuss the correctness of this statement in four points.

11. Explain six measures for minimizing the environmental degradation caused by extractions of metal.